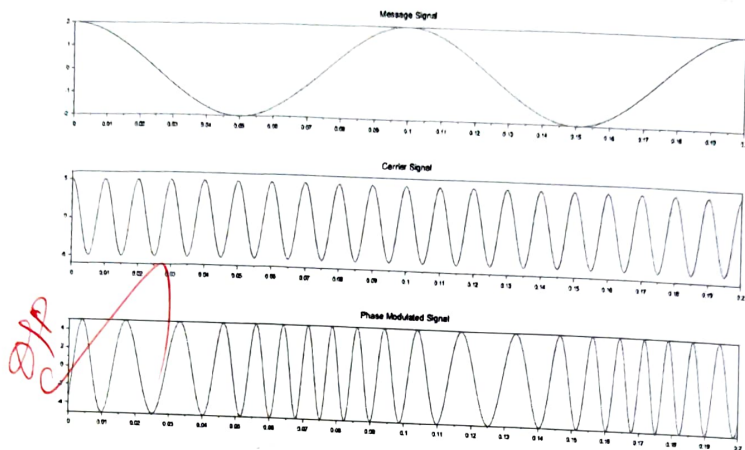




scilab program & Implementation (20)	Waveforms and tabulation (20)	calculation and result (20)	viva- voce (20)	Result and observation (20)

8/8/26

exp 5

PHASE MODULATION:-

AIM:-

To generate and simulate phase modulation in Scilab and calculate

EQUIPMENT REQUIRED:-

Scilab experiment.

CODE:-

```
clc;
```

```
clear;
```

```
clf;
```

```
Ac=5;
```

```
Am=3;
```

```
fc=100;
```

```
fm=10;
```

```
kp=2;
```

```
fs=5000;
```

```
t = 0:1/fs:0.2;
```

```
m = Am * cos(2 * %pi * fm * t);
```

```
c = Ac * cos(2 * %pi * fc * t);
```

```
pm = Ac * cos(2 * %pi * fc * t + kp * m);
```

```
subplot(3,1,1)
```

```
plot(t,m)
```

```
title("message signal")
```

S.no	Particular	Theory		Practical	
		Amplitude	frequency	Amplitude time	frequency
1.	Message signal	3	10	$T = X_2 - X_1$ $= 0.15 - 0.05$ $= 0.1s$	$F = \frac{1}{T} \Rightarrow \frac{1}{0.1}$ $\Rightarrow 10Hz$
2.	Carrier signal	5	1000	$T = X_2 - X_1$ $= 0.15 - 0.05$ $= 0.10s$	$F = \frac{1}{T} \Rightarrow \frac{1}{0.01}$ $\Rightarrow 100Hz$
3.	Phase modulation signal	—	—	$T_{min} = 0.078 - 0.071$ $= 0.007s$ $0.000947s$ $T_{max} = 0.133 - 0.117$ $\Rightarrow 0.016s$ $0.000100s$ $0.000943s$	$F_{max} = \frac{1}{T_{min}}$ $\approx 1060Hz$ $= 142.86Hz$ ≈ 940 $F_{min} = \frac{1}{T_{max}}$ $\approx 225Hz$ ≈ 1060 $F_{min} = 1 \approx 940Hz$ max

subplot (3,1,2)

plot (t, c)

title ("Carrier Signal")

subplot (3,1,3)

plot (t, pm)

title ("Phase modulated signal").

CALCULATION:-

$$\Delta f = A_m k_p f_m$$
$$= 2 \times 3 \times 10$$

$$\Delta f = 60 \text{ Hz}$$

$$f_{\min} = f_c - \Delta f$$
$$= 1000 - 60$$

$$f_{\min} = 940 \text{ Hz}$$

$$f_{\max} = f_c + \Delta f$$
$$= 1000 + 60$$

$$f_{\max} = 1060 \text{ Hz}$$

Scilab program & implementation (20)	Waveform and tabulation (20)	Calculation and result (20)	viva- voce (20)	Result and observation (20)
20	20	20	15	20 95

RESULT:-

Phase modulation is generated and simulated in Scilab.

est